REVISITING MODEL ORGANISMS FOR SANDBAGGING

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ABSTRACT

Malicious model developers may deliberately induce language models to underperform on dangerous tasks to evade regulation, a behavior known as *sandbagging*. Prior work has proposed model organisms that simulate sandbagging to evaluate techniques for eliciting such behavior. However, naïve model organisms can underestimate sandbagging risk and overestimate the effectiveness of elicitation techniques. Moreover, evaluation in mismatched threat models (as in prior work) can lead to a false sense of effectiveness of elicitation techniques. Thus, systematically designing *worst-case model organisms under appropriate threat models* is essential for establishing strong baselines and properly evaluating elicitation techniques. Yet, this remains unexplored in the literature.

We systematically explore the design of worst-case model organisms by identifying the following requirements: (i) *effective* (sandbags on dangerous tasks), (ii) *utility-preserving* (maintains performance on other tasks), and (iii) *robust* (resists existing elicitation techniques). Our empirical evaluation shows that existing model organisms are vulnerable to simple elicitation techniques and thus, do not represent the worst-case risk. We then propose three model organisms to demonstrate systematic worst-case design: two for black-box settings (explicitly optimized against blackbox elicitation like adversarial prompts), and one for white-box settings (optimized to resist whitebox elicitation such as fine-tuning, activation probing, pruning, and fuzzing). We show that our proposed model organisms are substantially more robust than prior work, providing stronger worst-case baselines for developing and evaluating future elicitation techniques.

1 INTRODUCTION

2 BACKGROUND AND RELATED WORK

3 PROBLEM STATEMENT

4 POSSUM: FRAMEWORK

5 EXPERIMENTAL SETUP

6 EVALUATION

7 DISCUSSION AND SUMMARY

REFERENCES

A APPENDIX

You may include other additional sections here.

Table 1: **Evaluation of Prior Works.** Effectiveness **??**, utility **??**, and robustness **??** across six sandbagging tasks and three Qwen2.5 sizes (7B, 14B, and 32B). For **??**, "w/o" is accuracy with the sandbagging disabled (unlocked), "w/" is accuracy with sandbagging, and Δ = w/o − w/ (higher = stronger). For **??** we report HellaSwag (HS) and CommonSenseQA (CSQA) with the same generate-and-parse utility readout as Table 2; task training, effectiveness, and final attack scoring remain forced-choice for prior-work fidelity. The fixed SysPr prompt uses seeded top-k decoding (three inference seeds); all other prior replicates are trained seeds. For **??**, we report AutoDAN attack accuracy (Jlbrk, lower = more robust).

Base model	WMDP-Bio						WMDP-Chem						WMDP-Cyber					
	Eff. ↑			Util. ↑		Rob. ↓	Eff. ↑			Util. ↑		Rob. ↓	Eff. ↑			Util. ↑		Rob. ↓
	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk
PWD																		
Qwen2.5-7B-Instruct	$11.2_{2.8}$	$69.7_{1.6}$	$58.5_{1.4}$	$74.0_{0.1}$	$81.0_{0.5}$	$96.9_{0.2}$	$25.7_{0.8}$	$43.4_{1.2}$	$17.7_{2.0}$	$74.3_{0.9}$	$81.1_{0.4}$	$92.0_{0.4}$	$17.5_{1.0}$	$48.0_{0.6}$	$30.6_{1.5}$	$76.4_{1.0}$	$80.0_{0.3}$	—
Qwen2.5-14B-Instruct	$9.9_{2.9}$	$74.0_{1.5}$	$64.1_{1.6}$	$83.2_{0.7}$	$80.8_{0.6}$	—	$20.5_{3.7}$	$52.6_{1.1}$	$32.1_{2.6}$	$79.4_{3.2}$	$78.7_{1.7}$	—	$15.0_{2.3}$	$46.8_{1.1}$	$31.7_{3.1}$	$81.3_{1.8}$	$80.5_{0.9}$	—
Qwen2.5-32B-Instruct	$7.2_{1.5}$	$81.2_{1.4}$	$74.1_{1.8}$	$87.4_{0.2}$	$85.6_{0.1}$	—	$18.9_{2.1}$	$52.6_{1.8}$	$33.7_{1.8}$	$88.0_{0.2}$	$86.2_{0.2}$	—	$12.4_{0.7}$	$60.8_{1.1}$	$48.5_{1.1}$	$87.1_{0.3}$	$85.4_{0.1}$	—
CktBr																		
Qwen2.5-7B-Instruct	$10.9_{2.7}$	$70.4_{1.2}$	$59.5_{3.1}$	$75.5_{0.3}$	$80.8_{0.2}$	—	$24.1_{1.8}$	$41.4_{1.4}$	$17.3_{1.8}$	$74.0_{1.4}$	$81.3_{0.5}$	—	$21.7_{5.2}$	$48.9_{0.8}$	$27.2_{4.7}$	$75.8_{0.7}$	$79.9_{0.2}$	—
Qwen2.5-14B-Instruct	$11.7_{2.6}$	$74.5_{0.6}$	$62.8_{3.2}$	$83.2_{1.0}$	$80.8_{0.9}$	—	$23.3_{1.8}$	$50.2_{1.4}$	$26.9_{2.9}$	$80.4_{2.5}$	$81.0_{1.3}$	—	$28.7_{2.1}$	$48.6_{1.8}$	$20.0_{3.3}$	$65.8_{12.0}$	$64.0_{10.8}$	—
Qwen2.5-32B-Instruct	$5.9_{0.2}$	$80.3_{1.1}$	$74.5_{0.9}$	$87.2_{0.4}$	$85.7_{0.2}$	—	$14.5_{1.2}$	$50.6_{0.0}$	$36.1_{1.2}$	$87.7_{0.1}$	$85.9_{0.2}$	—	$16.0_{2.6}$	$55.6_{1.5}$	$39.5_{2.2}$	$87.1_{0.3}$	$85.4_{0.3}$	—
SysPr																		
Qwen2.5-7B-Instruct	$63.0_{0.9}$	$66.0_{0.0}$	$3.0_{0.9}$	$73.8_{0.0}$	$76.5_{0.1}$	—	$41.8_{2.1}$	$43.4_{0.0}$	$1.6_{2.1}$	$73.7_{0.1}$	$76.6_{0.2}$	—	$45.4_{1.1}$	$46.1_{0.0}$	$0.7_{1.1}$	$73.6_{0.0}$	$76.4_{0.2}$	—
Qwen2.5-14B-Instruct	$74.9_{0.3}$	$77.3_{0.0}$	$2.5_{0.3}$	$53.8_{0.1}$	$72.3_{0.2}$	—	$43.8_{1.4}$	$47.0_{0.0}$	$3.2_{1.4}$	$54.3_{0.2}$	$72.3_{0.0}$	—	$53.7_{0.4}$	$47.9_{0.0}$	$-5.8_{0.4}$	$54.0_{0.2}$	$72.5_{0.4}$	—
Qwen2.5-32B-Instruct	$78.9_{0.2}$	$79.7_{0.0}$	$0.8_{0.2}$	$87.8_{0.1}$	$84.9_{0.1}$	$96.5_{0.8}$	$52.2_{2.8}$	$51.8_{0.0}$	$-0.4_{2.8}$	$87.8_{0.1}$	$84.6_{0.2}$	—	$54.2_{1.5}$	$51.1_{0.0}$	$-3.1_{1.5}$	$87.8_{0.0}$	$84.7_{0.2}$	—

Base model	MMLU-Prof						MMLU-Stem						MMLU-HSS					
	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk
PWD																		
Qwen2.5-7B-Instruct	$10.2_{0.6}$	$55.4_{0.9}$	$45.2_{0.3}$	$76.3_{1.3}$	$79.9_{0.3}$	$97.6_{0.1}$	$15.0_{1.8}$	$57.1_{1.0}$	$42.2_{1.0}$	$73.9_{1.6}$	$80.9_{0.1}$	$94.6_{1.0}$	$13.5_{0.8}$	$54.0_{0.9}$	$40.4_{1.7}$	$76.3_{0.2}$	$80.9_{1.1}$	—
Qwen2.5-14B-Instruct	$14.2_{2.2}$	$62.2_{1.4}$	$48.0_{2.7}$	$84.8_{0.2}$	$81.3_{0.3}$	—	$11.6_{0.4}$	$62.1_{0.5}$	$50.6_{0.8}$	$82.5_{1.9}$	$81.2_{1.2}$	—	$10.0_{2.2}$	$75.7_{2.6}$	$65.8_{0.9}$	$83.9_{0.7}$	$81.7_{0.1}$	—
Qwen2.5-32B-Instruct	$10.7_{1.2}$	$67.7_{1.1}$	$57.0_{2.2}$	$88.4_{0.3}$	$86.0_{0.2}$	—	$19.3_{10.1}$	$67.1_{3.4}$	$47.8_{12.8}$	$87.5_{0.3}$	$85.9_{0.3}$	—	$14.0_{1.0}$	$81.6_{0.8}$	$67.6_{0.9}$	$88.4_{0.1}$	$86.1_{0.1}$	—
CktBr																		
Qwen2.5-7B-Instruct	$44.3_{8.8}$	$54.5_{0.5}$	$10.3_{8.3}$	$77.1_{1.4}$	$80.1_{0.5}$	—	$19.7_{0.8}$	$56.9_{0.2}$	$37.2_{0.6}$	$73.8_{1.8}$	$80.4_{0.7}$	—	$28.2_{5.4}$	$52.4_{1.4}$	$24.3_{4.3}$	$74.8_{0.9}$	$80.5_{0.7}$	—
Qwen2.5-14B-Instruct	$36.8_{12.3}$	$62.2_{1.0}$	$25.4_{11.3}$	$82.6_{1.3}$	$81.1_{0.4}$	—	$12.5_{1.9}$	$57.4_{1.6}$	$44.9_{3.5}$	$82.6_{1.0}$	$81.6_{0.3}$	—	$38.4_{8.9}$	$73.3_{2.5}$	$34.9_{10.5}$	$56.3_{28.2}$	$56.0_{24.4}$	—
Qwen2.5-32B-Instruct	$11.1_{2.1}$	$67.6_{0.4}$	$56.5_{1.7}$	$87.5_{0.4}$	$85.8_{0.3}$	—	$9.8_{1.0}$	$67.3_{2.2}$	$57.6_{1.9}$	$87.2_{0.3}$	$85.5_{0.1}$	—	$11.4_{2.4}$	$80.3_{1.0}$	$68.9_{2.6}$	$88.3_{0.2}$	$86.4_{0.2}$	—
SysPr																		
Qwen2.5-7B-Instruct	$52.8_{0.4}$	$52.9_{0.0}$	$0.1_{0.4}$	$77.2_{0.1}$	$79.5_{0.1}$	—	$51.2_{1.3}$	$56.5_{0.0}$	$5.2_{1.3}$	$74.9_{0.1}$	$79.4_{0.1}$	—	$54.7_{0.3}$	$48.8_{0.0}$	$-5.9_{0.3}$	$72.9_{0.1}$	$79.4_{0.1}$	—
Qwen2.5-14B-Instruct	$57.1_{0.6}$	$61.5_{0.0}$	$4.5_{0.6}$	$66.9_{0.1}$	$70.9_{0.3}$	—	$61.2_{0.4}$	$53.7_{0.0}$	$-7.5_{0.4}$	$53.2_{0.2}$	$73.0_{0.3}$	—	$72.4_{0.4}$	$71.7_{0.0}$	$-0.6_{0.4}$	$64.0_{0.1}$	$76.4_{0.0}$	—
Qwen2.5-32B-Instruct	$49.6_{0.4}$	$67.3_{0.0}$	$17.7_{0.4}$	$57.2_{0.1}$	$45.8_{0.3}$	—	$46.3_{1.4}$	$61.9_{0.0}$	$15.6_{1.4}$	$53.0_{0.1}$	$35.7_{0.3}$	—	$40.4_{0.6}$	$74.7_{0.0}$	$34.2_{0.6}$	$67.8_{0.1}$	$55.9_{0.3}$	—

Table 2: **Evaluation of Proposed Model Organisms** across six sandbagging tasks and three Qwen2.5 sizes. **Effectiveness ??**: sandbag-task accuracy without (w/o) and with (w/) the induced behavior, Δ = w/o − w/ (higher = stronger). **Utility ??**: HellaSwag (HS) and CommonSenseQA (CSQA). **Robustness ??**: maximum blackbox jailbreak accuracy (Jlbrk, lower = more robust). Cells are shaded vs the matched prior-work baseline (Table 1) by ± standard-error bars over seeds: green = ours statistically better, amber = similar, red = worse; no red in an Eff. column confirms the lock is at least as strong.

Base model	WMDP-Bio						WMDP-Chem						WMDP-Cyber					
	Eff. ↑			Util. ↑		Rob. ↓	Eff. ↑			Util. ↑		Rob. ↓	Eff. ↑			Util. ↑		Rob. ↓
	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk
RobPWD																		
Qwen2.5-7B-Instruct	0.0	67.2	67.2	85.0	81.4	0.0	$2.4_{2.4}$	$44.6_{1.2}$	$42.2_{1.2}$	$81.8_{0.5}$	$80.3_{0.5}$	$0.0_{0.0}$	$0.0_{0.0}$	$49.0_{0.2}$	$49.0_{0.2}$	$80.5_{3.4}$	$79.6_{1.3}$	$0.0_{0.0}$
Qwen2.5-14B-Instruct	0.0	75.8	75.8	86.5	70.4	0.0	—	—	—	—	—	—	—	—	—	—	—	—
Qwen2.5-32B-Instruct	0.0	63.3	63.3	88.8	84.8	0.0	—	—	—	—	—	—	—	—	—	—	—	—
RobAdapt																		
Qwen2.5-7B-Instruct	$10.4_{7.0}$	$69.1_{0.0}$	$58.7_{7.0}$	$72.8_{4.4}$	$65.9_{8.7}$	$4.2_{2.1}$	$18.1_{9.1}$	$41.0_{0.0}$	$22.9_{9.1}$	$73.0_{2.0}$	$73.6_{4.0}$	$16.7_{12.3}$	$0.9_{0.5}$	$45.6_{0.0}$	$44.7_{0.5}$	$64.0_{14.9}$	$68.5_{8.6}$	$2.1_{2.1}$
Qwen2.5-14B-Instruct	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Qwen2.5-32B-Instruct	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
RoBCoTDstl																		
Qwen2.5-7B-Instruct	$0.0_{0.0}$	$59.8_{2.6}$	$59.8_{2.6}$	$80.7_{0.4}$	$74.0_{1.3}$	$0.0_{0.0}$	$2.4_{2.4}$	$38.6_{1.8}$	$36.1_{0.7}$	$79.3_{3.8}$	$57.0_{17.2}$	$9.4_{9.4}$	$10.6_{10.5}$	$59.6_{1.2}$	$49.0_{10.5}$	$81.8_{0.9}$	$77.9_{0.3}$	$0.0_{0.0}$
Qwen2.5-14B-Instruct	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Qwen2.5-32B-Instruct	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Base model	MMLU-Prof						MMLU-Stem						MMLU-HSS					
	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk	w/	w/o	Δ	HS	CSQA	Jlbrk
RobPWD																		
Qwen2.5-7B-Instruct	$0.2_{0.2}$	$45.4_{9.4}$	$45.2_{9.3}$	$74.8_{5.3}$	$78.4_{0.7}$	$5.2_{5.2}$	$8.6_{8.6}$	$58.6_{0.9}$	$50.0_{9.5}$	$81.5_{1.5}$	$79.8_{1.3}$	$1.0_{1.0}$	$0.4_{0.4}$	$49.3_{5.5}$	$48.9_{5.7}$	$77.2_{2.6}$	$79.8_{0.9}$	$0.0_{0.0}$
Qwen2.5-14B-Instruct	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Qwen2.5-32B-Instruct	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
RobAdapt																		
Qwen2.5-7B-Instruct	$14.4_{8.9}$	$56.1_{0.0}$	$41.7_{8.9}$	$74.6_{2.0}$	$73.7_{1.3}$	$26.0_{14.7}$	$46.8_{11.4}$	$62.2_{0.0}$	$15.3_{11.4}$	$68.4_{1.2}$	$75.9_{1.8}$	$49.0_{19.9}$	$5.9_{3.1}$	$51.6_{0.0}$	$45.7_{3.1}$	$66.6_{9.0}$	$53.9_{20.7}$	$12.5_{11.0}$
Qwen2.5-14B-Instruct	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Qwen2.5-32B-Instruct	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
RoBCoTDstl																		
Qwen2.5-7B-Instruct	$56.3_{3.2}$	$61.2_{3.4}$	$4.9_{6.5}$	$73.0_{10.9}$	$75.9_{1.9}$	$18.8_{9.5}$	$34.2_{17.2}$	$54.5_{1.8}$	$20.3_{16.5}$	$81.8_{0.8}$	$73.0_{4.6}$	$0.0_{0.0}$	$58.2_{5.3}$	$60.1_{3.9}$	$2.0_{1.4}$	$80.2_{1.2}$	$77.4_{1.5}$	$0.0_{0.0}$
Qwen2.5-14B-Instruct	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Qwen2.5-32B-Instruct	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Table 3: **Robustness ?? of `RoBCoTDstl`.** Attack accuracy of whitebox detection/elicitation methods (SFT, DPO, AntRef, Steering, LAT, Pruning, Fuzzing, Probing, FT + Probing) across six sandbagging tasks and three Qwen2.5 sizes; lower = more robust.

Base model	SFT	DPO	AntRef	Steering	LAT	Pruning	Fuzzing	Probing	FT + Probing
WMDP-Bio									
Qwen2.5-7B-Instruct	—	—	—	—	—	*n/a*	—	$99.9_{0.1}$	—
Qwen2.5-32B-Instruct	—	—	—	—	—	*n/a*	—	—	—
Qwen2.5-72B-Instruct	—	—	—	—	—	*n/a*	—	—	—
WMDP-Chem									
Qwen2.5-7B-Instruct	—	—	—	—	—	*n/a*	—	$100.0_{0.0}$	—
Qwen2.5-32B-Instruct	—	—	—	—	—	*n/a*	—	—	—
Qwen2.5-72B-Instruct	—	—	—	—	—	*n/a*	—	—	—
WMDP-Cyber									
Qwen2.5-7B-Instruct	—	—	—	—	—	*n/a*	—	—	—
Qwen2.5-32B-Instruct	—	—	—	—	—	*n/a*	—	—	—
Qwen2.5-72B-Instruct	—	—	—	—	—	*n/a*	—	—	—
MMLU-Prof									
Qwen2.5-7B-Instruct	—	—	—	—	—	*n/a*	—	—	—
Qwen2.5-32B-Instruct	—	—	—	—	—	*n/a*	—	—	—
Qwen2.5-72B-Instruct	—	—	—	—	—	*n/a*	—	—	—
MMLU-Stem									
Qwen2.5-7B-Instruct	—	—	—	—	—	*n/a*	—	—	—
Qwen2.5-32B-Instruct	—	—	—	—	—	*n/a*	—	—	—
Qwen2.5-72B-Instruct	—	—	—	—	—	*n/a*	—	—	—
MMLU-HSS									
Qwen2.5-7B-Instruct	—	—	—	—	—	*n/a*	—	—	—
Qwen2.5-32B-Instruct	—	—	—	—	—	*n/a*	—	—	—
Qwen2.5-72B-Instruct	—	—	—	—	—	*n/a*	—	—	—